EX 9	Google Maps Integration

Aim

To create a location-aware app using Google Maps.

Definitions

Google Maps

Google Maps is a comprehensive web mapping platform and consumer application developed by Google, offering satellite imagery, street maps, 360° panoramic views (Street View), real-time traffic conditions, and route planning for driving, biking, walking, and public transit. Available on mobile and desktop, it helps users explore the world, locate businesses, and navigate.

Procedure

1. Open Notepad, type the following codes and save file as map.html in the desktop.

map.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Google Map</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        html, body {
            margin: 0;
            padding: 0;
            height: 100%;
        }

        #map {
            height: 100%;
            width: 100%;
        }
    </style>
</head>

<body>

<div id="map"></div>

<script
src="https://maps.googleapis.com/maps/api/js?key=AIzaSyDKNgqUxDwM9sglnjjt7Ncr4GVYLG2VlFQ"></script>

<script>
function initMap() {
    var loc = { lat: 13.0827, lng: 80.2707 };

    var map = new google.maps.Map(document.getElementById("map"), {
        zoom: 12,
        center: loc
    });

    new google.maps.Marker({
        position: loc,
        map: map
    });
}
```

```
window.onload = initMap;
</script>

</body>
</html>
```

2. Open New file in Notepad, type the following codes and save file as MainActivity.java in the desktop.

MainActivity.java

```
import java.io.*;
import java.net.*;

public class SimpleServer {

    public static void main(String[] args) throws Exception {

        ServerSocket serverSocket = new ServerSocket(8080);
        System.out.println("Server started at http://localhost:8080");

        while (true) {
            Socket socket = serverSocket.accept();

            BufferedReader in = new BufferedReader(
                    new InputStreamReader(socket.getInputStream()));

            OutputStream out = socket.getOutputStream();

            // Read request (not used)
            while (in.ready()) {
                in.readLine();
            }

            // Load HTML file
            File file = new File("map.html");
            BufferedReader fileReader = new BufferedReader(new FileReader(file));

            String line;
            StringBuilder content = new StringBuilder();

            while ((line = fileReader.readLine()) != null) {
                content.append(line);
            }

            // Send HTTP response
            String response = "HTTP/1.1 200 OK\r\n" +
                    "Content-Type: text/html\r\n\r\n" +
                    content.toString();

            out.write(response.getBytes());
```

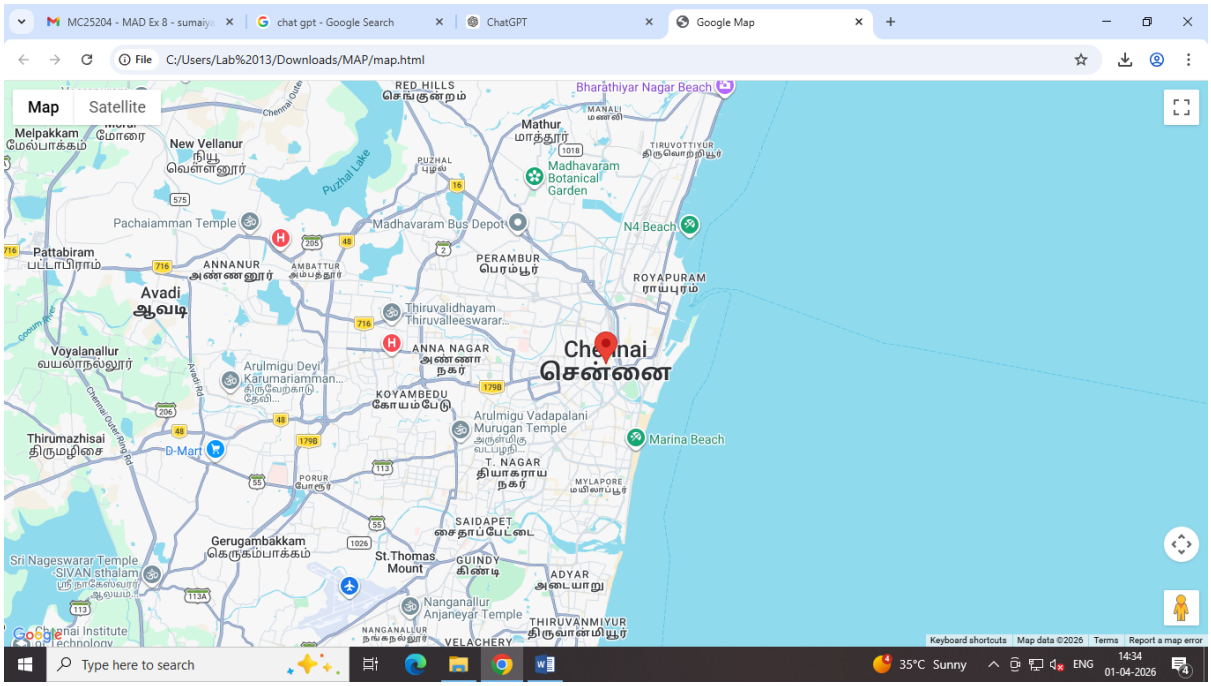
```
            out.flush();

            socket.close();
        }
    }
}
```

3. Follow the steps to generate API key for JavaScript (Web):
    - Go to: https://console.cloud.google.com/
    - Sign in with your Google account
    - Click **Select Project**
    - Click **New Project**
    - Enter a project name
    - Click **Create**
    - Select the newly created project

    - Go to **APIs & Services → Library**
    - Search for **Maps JavaScript API / Maps SDK for Android**
    - Click it and press **Enable**
    - Go to **APIs & Services → Credentials**
    - Click **+ Create Credentials**
    - Choose **API Key**
    - Your API key will be created instantly
    - Copy and use the generated API key in your Coding
4. Open map.html in the web browser to see the output

Output



Result

Thus, a location-aware app has been created using Google Maps.

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